ECON 4555: Part III, Question 4

Summary Statistics in Levels

2025 Q2: April–June

The sample is the validated Part II baseline: one eligible ticketing carrier per unordered airport pair, for a total of 7,499 observations. All continuous regression variables below are reported in levels.

Table 1: Dependent and independent variables in the planned fare regression

Variable	Units	N	Mean	Std. dev.	Minimum	Maximum
20th-percentile fare	Dollars	7,499	174.97	70.28	20.00	953.50
50th-percentile fare	Dollars	7,499	243.73	93.31	23.50	1,708.00
80th-percentile fare	Dollars	7,499	348.04	127.96	32.00	1,982.00
Nonstop O&D distance	Miles	7,499	1,233.25	815.35	41.00	6,089.00
Load factor	Ratio (0–1)	7,499	0.8152	0.0699	0.1495	0.9537
Circuity	Ratio	7,499	1.0960	0.1610	0.9973	2.6677
Coupon-stop proxy	Coupon points	7,499	0.6309	0.4665	0.0000	1.0000
Route HHI	Index (0–1)	7,499	0.5727	0.2466	0.1596	1.0000
Route share	Share (0–1)	7,499	0.5961	0.2844	0.1001	1.0000
Airport-origin HHI	Index (0–1)	7,499	0.3491	0.1427	0.1476	1.0000
Airport-origin share	Share (0–1)	7,499	0.3480	0.1791	0.0083	1.0000

Notes. Statistics are unweighted across the 7,499 baseline carrier–route observations; the standard deviation uses the sample denominator $N-1$. Missing values are omitted separately for each variable. All 11 reported variables are complete, so no observations are removed from the baseline for this table. Categorical fixed-effect identifiers are not summarized as continuous variables. No regressions have been estimated.

Fares are passenger-weighted within-carrier/route percentiles of allocated directional DB1B Market fares, in nominal dollars. The table summarizes those percentile fares across observations; it does not recalculate pooled passenger fare percentiles. Distance is nonstop origin–destination mileage. Load factor is a passenger-weighted measure of quarterly operating-segment passenger/seat ratios, with coupon-distance weights within journeys. Circuity is flown mileage divided by nonstop mileage. Slight values below one in the source distance measures are retained.

Stops limitation. The coupon-stop proxy is the passenger-weighted mean of market coupons minus one. It counts intermediate coupon points, cannot distinguish on-plane stops from plane changes, and can miss stops within a coupon. True on-plane stops are unavailable in all 7,499 observations and are not treated as zero or used to delete the entire sample.

Route and airport-origin shares and HHIs are measured on a **0–1 scale**, not as percentages or on the 0–10,000 HHI scale. Airport-origin variables combine endpoint ticket-origination measures using weights based on true ticket origins. Route and airport share denominators were constructed using the broader quarter before regression-sample restrictions.

Main patterns. Mean 20th-, 50th-, and 80th-percentile fares are \$174.97, \$243.73, and \$348.04. Mean distance is 1,233 miles and mean load factor is 0.8152 (81.52%). Mean circuity of 1.0960 implies about 9.6% more flown mileage than nonstop mileage on average. Mean route HHI (0.5727) exceeds mean airport-origin HHI (0.3491); the selected carrier's mean route share is 0.5961. These patterns describe the selected baseline and do not establish causal effects.

Source: Validated BTS DB1B Market, Ticket and Coupon, and T-100 Domestic Segment data for 2025 Q2; Part II baseline file `q4_baseline_one_carrier_per_route.csv`.